systemDiscovery

Service Description

Service ID: *"systemDiscovery"*

Abstract
This document describes the `systemDiscovery` service, version 1.0.0, produced by the ServiceRegistry core system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `systemDiscovery` service, which provides registration and lookup of the application, support and core systems of the local cloud, with their addresses, version, metadata and hosting device. It is produced by the ServiceRegistry system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Corresponds to the system part of the Arrowhead 4.x ServiceRegistry `register`/`unregister` operations and the 4.4 SystemRegistry.

2.2 How This Service Is Meant to Be Used

A system registers itself with `register` before it registers services; the system name is taken from its authenticated identity, not from the request. `revoke` removes the requester and, with it, its service instances. `lookup` finds systems by name, address, version, device or metadata; it is used by ConsumerAuthorization to evaluate metadata-based policies.

2.3 Important Delimitations and Dependencies

A system can only register and revoke itself. Registering another system requires `serviceRegistry Management`. The service does not authenticate systems; the identity comes from the authentication policy of the ServiceRegistry (certificate, identity token or declared name).

3 Service Interface

This section describes the interfaces to the `systemDiscovery` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `systemDiscovery` service and its operations.

The following interface operations are available.

3.1 operation register (SystemRegisterRequest) : SystemResponse

Registers the requesting system (name from its identity) with addresses, version, metadata and an optional device. Returns 201 for a new entry, 200 if an identical entry exists; an existing system with different data is refused (400).

3.2 operation lookup (SystemLookupRequest) : SystemListResponse

Returns the systems matching all given filters. With `verbose=true` (and `discovery.verbose` enabled) the device details are included.

Optional parameter: `verbose`.

3.3 operation revoke ()

Removes the requesting system and its service instances. Returns 200 if removed, 204 if it was not registered.

4 Information Model

Here, all data objects that can be part of the `systemDiscovery` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.8, which are used to represent things like hashes and identifiers.

4.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

4.2 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.3 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

4.4 struct SystemListResponse

List of systems.

Field	Type	Description
entries	List<SystemResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

4.5 struct SystemLookupRequest

Filter for systems; all fields optional, combined with AND.

Field	Type	Description
systemNames	List<Name>	System names to match.
addresses	List<String>	Network addresses of the entity.
addressType	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
metadataRequirementList	List<MetadataRequirement>	Metadata requirements; each map is one alternative.
versions	List<Version>	System versions to match.
deviceNames	List<Name>	Device names to match.

4.6 struct SystemRegisterRequest

Registers the requesting system; the name is taken from its authenticated identity.

Field	Type	Description
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<String>	Network addresses of the entity.
deviceName	Name	Name of the device hosting the system.

4.7 struct SystemResponse

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponse	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.8 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, serviceregistry/documentation/v5.3.0/ServiceRegistry-v5.3.0-SysD.pdf.

[2] ——, “systemDiscovery HTTP/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-systemDiscovery-HTTP-IDD.pdf.

[3] ——, “systemDiscovery MQTT/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.3.0, 2026, serviceRegistry-v5.3.0-systemDiscovery-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	